



CLDV6212



Dillon Rinkwest-

Table of Contents

Section A.....	2
YouTube Link.....	4
GitHub Link.....	4
Section B.....	5
Section C	6
Azure Blob Storage	6
Azure Queue Storage.....	6
Azure File Share	6
Azure	6
Azure SQL Database	6
Azure App Service	7
Azure Table Storage	7
Section D	8
Alternative Azure Technologies	8
References	9
Disclosure of AI Usage in my Assessment	11

Section A

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags **Review + create**

Product details

SQL database by Microsoft
[Terms of use](#) | [Privacy policy](#)

Estimated cost per month
7.51 USD

Cost summary

Basic (Basic)	1.50
Cost per DTU (in USD)	x 5
DTUs selected	
ESTIMATED COST / MONTH	7.51 USD

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms of Use](#).

Basics

Subscription	ADVTECH-Tertiary Varsity College
Resource group	AZ-JHB-RSG-VCVCCN-ST10439147-TER
Region	South Africa North
Database name	st10439147-DB
Server	(new) osh6212-st10439147
Authentication method	SQL and Microsoft Entra authentication
Server admin login	st10439147
Microsoft Entra Admin	ST10439147@vconnect.edu.za
Compute + storage	Basic 2 GB storage
Backup storage redundancy	Locally-redundant backup storage

Networking

Allow Azure services and resources to access this server	Yes
--	-----

Create **< Previous** [Download a template for automation](#)

Backup storage redundancy Locally-redundant backup storage

Networking

Allow Azure services and resources to access this server	Yes
Add current client IP address	Yes
Private endpoint	None
Minimum TLS version	1.2
Connection Policy	Default

Security

Identity	Not enabled
Transparent data encryption (Server level)	Service-managed key selected
Database level customer-managed key	Not configured
Database level user assigned managed identity	Not configured
Advanced data security	Not now
Always Encrypted with secure enclaves	Not configured
Sql Ledger(Database)	Disabled
Digest Storage	Disabled

Additional settings

Use existing data	Blank
Collation	SQL_Latin1_General_CP1_CI_AS
Maintenance window	System default (8pm to 9am)

Tags

Create **< Previous** [Download a template for automation](#)

Microsoft.SQlDatabase.newDatabaseNewServer_12e40724a1284793873ff | Overview

Deployment

Search

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.SQlDatabase.newDatabaseNewServer_12e40724a1284793873ff

Subscription : ADVTECH-Tertiary College

Resource group : AZ-JHB-R35-VCMCDN-ST10439147-TER

Start time : 11/10/2025, 2:37:00 PM

Correlation ID : 13316ba1-2768-4d78-8777-59480623d271

Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Find an Azure expert >

st10439147-DB (cldv6212-st10439147/st10439147-DB) | Query editor (preview)

SQL database

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource Visualizer

Tables

Views

Stored Procedures

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Query 1

Run

Cancel query

Save query

Export data as

Show only Editor

```
1 -- Create Users Table for authentication
2 CREATE TABLE Users (
3     UserId INT PRIMARY KEY IDENTITY(1,1),
4     Email NVARCHAR(100) UNIQUE NOT NULL,
5     PasswordHash NVARCHAR(256) NOT NULL,
6     Role NVARCHAR(20) NOT NULL, -- 'Customer' or 'Admin'
7     CustomerId NVARCHAR(100) NOT NULL, -- Links to Azure Table Storage Customer RowKey
8     IsActive BIT NOT NULL DEFAULT 1,
9     CreatedDate DATETIME NOT NULL DEFAULT GETUTCDATE(),
10    LastLoginDate DATETIME NULL
11 );
12
13 -- Create index on Email for faster lookups
14 CREATE INDEX IX_Users_Email ON Users(email);
```

Results

Messages

Search to filter items...

The top screenshot shows the Microsoft Azure portal interface for the 'st10439147-DB' database. The 'Query editor (preview)' is open, displaying a SQL query that inserts a new user into the 'Users' table. The query is as follows:

```

27 SELECT HASHBYTES('SHA2_256', 'Admin1231') AS H
28 ) AS X;
29
30 -- Password hash is SHA256 of 'Admin1231'
31 INSERT INTO Users (Email, PasswordHash, Role, CustomerId, IsActive, CreatedDate)
32 VALUES ('admin@abccretail.com',
33         'PwPz2rMc08QmClyt5ccRjpdvuuTduqHEBRVRAoSe+',
34         'Admin',
35         NULL,
36         1,
37         GETUTCDATE());
38
39 -- Verify the Table
40 SELECT * FROM Users;

```

The bottom screenshot shows the same portal interface, but the query editor now displays the results of a query that selects the top 1000 users from the 'Users' table. The query is as follows:

```

1 SELECT TOP (1000) * FROM [dbo].[Users]

```

The results of the query are displayed in a table with the following columns: UserId, Email, PasswordHash, Role, CustomerId, IsActive, CreatedDate, and LastLoginDate. The table contains 4 rows of data.

UserId	Email	PasswordHash	Role	CustomerId	IsActive	CreatedDate	LastLoginDate
1	admin@abccretail.com	PwPz2rMc08QmClyt5ccRjpdvuuTduqHEBRVRAoSe+	Admin	9dbcc555-c111-4f85-9152-...	True	2025-11-12T15:54:03.2170...	2025-11-14T06:01:11.1135...
2	mic@yahoo.com	jHfW0oaM3bUaBqR9a...	Customer	9dbcc555-c111-4f85-9152-...	True	2025-11-13T06:01:37.1470...	2025-11-13T18:49:15.0670...
3	dylan@gmail.com	b5ry62L3vuuHm+E2RUa...	Customer	8361629-afce-4f1d-bed3-...	True	2025-11-13T21:16:32.1370...	2025-11-14T06:18:09.9100...
4	mic@joe.com	QhK3MmN1kpm7OpQ54L...	Customer	de9d38cc-e50c-4a6b-85e7-...	True	2025-11-13T22:56:49.0700...	2025-11-13T22:57:04.2830...

YouTube Link

<https://youtu.be/P-oUV2yOKHA>

GitHub Link

https://github.com/VCCT-CLDV6212-2025-G1/ST10439147_CLDV6212_POE/tree/Final

Section B

The ABC Retail cloud solution utilises several Azure services, each of which has been selected for its suitability to meet the specific functional and architectural needs of the application. The primary service used is the **Azure App Service**, which serves as the hosting environment for the web application. The Azure App Service operates as a Platform-as-a-Service (PaaS) solution. This is because it allows the system to benefit from automatic scaling, continuous deployment, integrated security, and simplified maintenance (Microsoft, 2025). Hence, this makes it the ideal choice for hosting the main interface of the online store. Complementing this is **Azure Functions**, which is another PaaS service that enables the serverless execution of backend logic, such as processing orders, handling queues, and managing file uploads (Kirvan & Bigelow, 2023). Azure Functions was selected due to its event-driven model, ability to scale automatically, cost-efficient consumption-based pricing, which ensures that the system uses computing resources only when required.

To store structured and relational data, such as customer and admin information for authentication, the solution uses **Azure SQL Database**. It is a fully managed relational database under the PaaS model. Azure SQL Database offers high availability, automated backups, geo-redundancy, and strong features, making it suitable for mission-critical transactional workloads (Microsoft, 2025). For semi-structured and non-relational data, the application utilises **Azure Table Storage**, which is a NoSQL key-value store that provides extremely fast, scalable, and cost-effective access to customer metadata and lightweight product data (Microsoft, 2025). Table Storage also follows a PaaS hosting model, offering schema flexibility and high throughput for frequently queried information.

In addition to these storage components, the solution uses **Azure Blob Storage** to store unstructured data such as product images and multimedia files (GeeksforGeeks, 2025). Blob Storage is also a PaaS model that provides basically unlimited scalability, global redundancy, and secure URL-based access, which makes it ideal for storing and retrieving large files required by the e-commerce system. For asynchronous message handling, the system uses **Azure Queue Storage**. Queue Storage is a PaaS-based messaging service that supports reliable communication between components (Microsoft, 2025). Queue Storage was also selected because it enables decoupling of application processes, ensuring smooth order handling even during high-volume traffic periods.

The final storage component is **Azure File Share**. File Share is used as a PaaS storage service to support the upload and management of contract documents and shared files. Azure File Share provides an SMB-based cloud file storage that can be mounted across multiple systems, making it suitable for administrative tasks and document-sharing functionalities required by the business (Awati & Boger, 2024). Working together, the seven Azure services represent a balanced set of PaaS-based technologies that maximise scalability, minimise maintenance overhead, and ensure the application meets ABC Retail's performance and reliability requirements.

Section C

The ABC Retail web application makes use of several Azure Services, each being selected to meet the specific business and technical requirements, such as scalability, cost-efficiency, and reliability. The following section provides the motivation for each Azure Service that was selected, explaining how it supports the functional goals of the system and contributes to overall system performance.

Azure Blob Storage

The first major functionality in the application is **product image management**, which uses **Azure Blob Storage**. Blob Storage was chosen because it provides a highly scalable and cost-effective way to store unstructured data such as images and multimedia files (LogicMonitor, 2025). It integrates easily with web applications and supports secure access through Shared Access Signatures, ensuring product images are quickly retrievable for display to customers (GeeksforGeeks, 2025). It is a pay-as-you-go model that allows the system to handle seasonal traffic surges without any manual intervention.

Azure Queue Storage

The second functionality involves transaction and order processing, which is handled by Azure Queue Storage. This service was selected due to it providing a simple, asynchronous messaging system that decouples components within the application, improving reliability and responsiveness (Mark, 2023). When a customer places an order, a message is written to a queue, ensuring that the order processing tasks can be handled sequentially without losing queue requests, even when the system is under heavy load. Queue Storage also integrates seamlessly with Azure Functions, allowing for automated back-end processing.

Azure File Share

The third functionality relates to files and document management, which is implemented by using Azure File Share. This service provides a managed SMB-based file storage solution that can be easily mounted to virtual machines or accessed via REST API (Awati & Boger, 2024). Azure File Share was selected because it offers compatibility with traditional file system operations while providing cloud scalability. This makes it ideal for storing contract documents and other administrative resources that multiple users or processes may need to access securely.

Azure Functions

The fourth functionality is the execution of serverless logic through Azure Functions. Azure Functions were chosen for their event-driven model, automatic scaling, and cost efficiency (Kirvan & Bigelow, 2023). They allow the application to execute background operations such as order confirmation, data uploads, and queue message handling without requiring continuous server allocation. This supports operational efficiency and ensures that resources are only consumed when the functions are triggered.

Azure SQL Database

The fifth functionality is data storage and analytics, which uses Azure SQL Database. This is a relational, fully managed Platform-as-a-Service (PaaS) solution that enables structured storage

for customer and user data. It was selected for its high availability, automated backup features, and built-in security mechanisms (Microsoft, 2025). Azure SQL Database also supports replication across regions, ensuring business continuity and improved read performance for geographically distributed customers.

Azure App Service

The sixth functionality in the ABC Retail solution is Azure App Service, which is used for hosting the web application. Azure App Service was selected because it provides reliable, fully managed environment for deploying web applications with minimal configuration. It supports continuous integration with GitHub, automatic scaling, and built-in monitoring, which simplifies deployment and maintenance (Microsoft, 2025). The combination of the App Service with other Azure components ensures that the solution is scalable, secure and easy to maintain.

Azure Table Storage

The seventh functionality in the ABC Retail solution involves the storage of customer profiles, product-related information, and order information, which is implemented using Azure Table Storage. This service was selected because it provides a highly scalable NoSQL key-value store that supports large volumes of structured, non-relational data (Microsoft, 2025). Azure Table Storage allows for fast read and write operations, making it well-suited for scenarios where data needs to be retrieved quickly, such as displaying customer profiles or product listings in real time. It offers a schema-less design, which enables flexible data modelling and easier adaptation as the application evolves (Microsoft, 2025). Table Storage also integrates seamlessly with Azure Functions and other services, enabling automated updates or event-driven processes when data changes occur. This is also one of the most cost-effective Azure data storage options, providing high availability with low latency while needing minimal maintenance. This makes Table Storage a good choice for storing lightweight, scalable, and frequently accessed data that supports the overall responsiveness of the ABC Retail web application.

Each Azure Service used in the ABC Retail solution was chosen to align with the system's operational goals. The combination of Blob Storage, Queue Storage, File Share, Azure Functions, SQL Database, App Service, and Table Storage ensures a cloud-native architecture that optimises costs, enhances reliability, and maintains flexibility for future expansion.

Section D

Alternative Azure Technologies

When looking through the current Azure services used within the ABC Retail solution, several alternative Azure technologies can be considered to enhance the scalability, reliability, and cost efficiency of the application.

The current implementation uses **Azure Blob Storage**, which is ideal for hosting static web content and multimedia files. A viable alternative would be **Azure Data Lake Storage Gen2**. It extends Blob Storage capabilities by offering a hierarchical namespace and easy integration with advanced analytics tools, such as Azure Synapse Analytics and Databricks (Microsoft, 2025). With this alternative, it would provide more improved performance and manageability for large datasets, especially if there is future expansion that includes predictive analytics or AI-driven product recommendations.

The second requirement revolves around message processing for orders and transactions. The current solution uses **Azure Queue Storage** for lightweight and reliable queueing. An alternative would be **Azure Service Bus**, which is a much more complex messaging architecture that supports topics, subscriptions, and guaranteed delivery. The use of Azure Service Bus would be beneficial for complex enterprise scenarios that require event-driven communication between multiple services (Microsoft, 2025). Although the Service Bus is more expensive and requires additional configuration, it offers greater scalability and reliability, which ends up reducing the risk of message loss during periods of high traffic.

The third requirement relates to the backend logic for the system. The current design implementation employs **Azure Functions**, which are highly scalable and cost-effective for event-driven workloads. A possible alternative would be **Azure App Service WebJobs**. It integrates directly with App Service hosting and allows continuous or scheduled execution with a persistent context. This approach would be preferable for long-running or background tasks that require greater control over the execution and scaling behaviour (Microsoft, 2025). Although WebJobs will require more funds compared to Functions, they provide more predictable runtime control.

The fourth requirement is the authentication system that uses the **Azure SQL Database** to store the user information in a relational format. A very strong alternative would be **Azure Cosmos DB** using the SQL API. Cosmos DB provides globally distributed, multi-model data storage with low-latency access across regions (Microsoft, 2024). This would enhance the performance for international users and improve disaster recovery capabilities. But the SQL Database is still better to use for workloads that depend on strict ACID (Atomicity, Consistency, Isolation, Durability) compliance and complex relational queries.

Each of these alternative Azure technologies is **Azure Data Lake Storage Gen2**, **Service Bus**, **App Service WebJobs**, and **Cosmos DB**. They offer potential improvements in specific areas of performance, scalability, and availability. The best choice will only be discovered based on ABC Retail's evolving business needs, data growth patterns, and cost considerations. Evaluating these alternatives shows the flexibility of the Azure cloud ecosystem and especially highlights how decisions for architecture can be customised to meet both current and future operational requirements.

References

- Awati, R. & Boger, T., 2024. *What is Microsoft Azure File Service?*. [Online]
Available at: <https://www.techtarget.com/searchwindowsserver/definition/Microsoft-Azure-File-Service>
[Accessed 14 November 2025].
- Deshmukh, A. S., 2025. *What is an Azure Service Bus?*. [Online]
Available at: <https://www.scholarhat.com/tutorial/azure/service-bus-architecture-components-features#:~:text=Azure%20Service%20Bus%20provides%20three,messages%20are%20stored%20until%20consumed.>
[Accessed October 2025].
- GeeksforGeeks, 2025. *Azure Blob Storage*. [Online]
Available at: <https://www.geeksforgeeks.org/cloud-computing/azure-blob-storage/>
[Accessed 14 November 2025].
- Imperium Dynamics, 2025. *Azure Service Bus*. [Online]
Available at: <https://imperiumdynamics.com/azure-service-bus-usecases-&-benefits#:~:text=Azure%20Service%20Bus%20elevates%20the,scalable%20and%20streamlined%20application%20frameworks.>
[Accessed October 2025].
- Kirvan, P. & Bigelow, S. J., 2023. *Microsoft Azure Functions*. [Online]
Available at: <https://www.techtarget.com/searchcloudcomputing/definition/Microsoft-Azure-Functions>
[Accessed 14 November 2025].
- Kurbegovic, E., 2024. *Understanding When to Use Azure Event Hub vs. Service Bus and Why*. [Online]
Available at: <https://medium.com/@emer.kurbegovic/understanding-when-to-use-azure-event-hub-vs-service-bus-and-why-534a20317ec3>
[Accessed October 2025].
- LogicMonitor, 2025. *What is Azure Blob Storage?*. [Online]
Available at: <https://www.logicmonitor.com/blog/what-is-azure-blob>
[Accessed 1 November 2025].
- M., 2023. *What Are Azure Queues and How Do They Work?*. [Online]
Available at: <https://www.smikar.com/what-are-azure-queues/>
[Accessed 13 November 2025].
- Microsoft, 2024. *Data modeling in Azure Cosmos DB*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/cosmos-db/nosql/modeling-data>
[Accessed 14 May 2025].
- Microsoft, 2025. *App Service overview*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/app-service/overview>
[Accessed 14 November 2025].

Microsoft, 2025. *Azure Data Lake Storage Gen2 overview in HDInsight*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/hdinsight/overview-data-lake-storage-gen2>
[Accessed 14 November 2025].

Microsoft, 2025. *Choose between Azure messaging services - Event Grid, Event Hubs, and Service Bus*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/service-bus-messaging/compare-messaging-services>
[Accessed October 2025].

Microsoft, 2025. *Introduction to Azure Data Lake Storage*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/storage/blobs/data-lake-storage-introduction>
[Accessed 14 November 2025].

Microsoft, 2025. *Introduction to Azure Storage*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/storage/common/storage-introduction>
[Accessed November 2025].

Microsoft, 2025. *Run background tasks with WebJobs*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/app-service/webjobs-create?tabs=windowscode>
[Accessed 14 November 2025].

Microsoft, 2025. *What is Azure Service Bus?*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/service-bus-messaging/service-bus-messaging-overview>
[Accessed October 2025].

Microsoft, 2025. *What is Azure SQL Database?*. [Online]
Available at: <https://learn.microsoft.com/en-us/azure/azure-sql/database/sql-database-paas-overview?view=azuresql>
[Accessed 14 November 2025].

Microsoft, 2025. *What is Azure Table storage?*. [Online]
Available at: <https://learn.microsoft.com/hu-hu/azure/storage/tables/table-storage-overview>
[Accessed 14 November 2025].

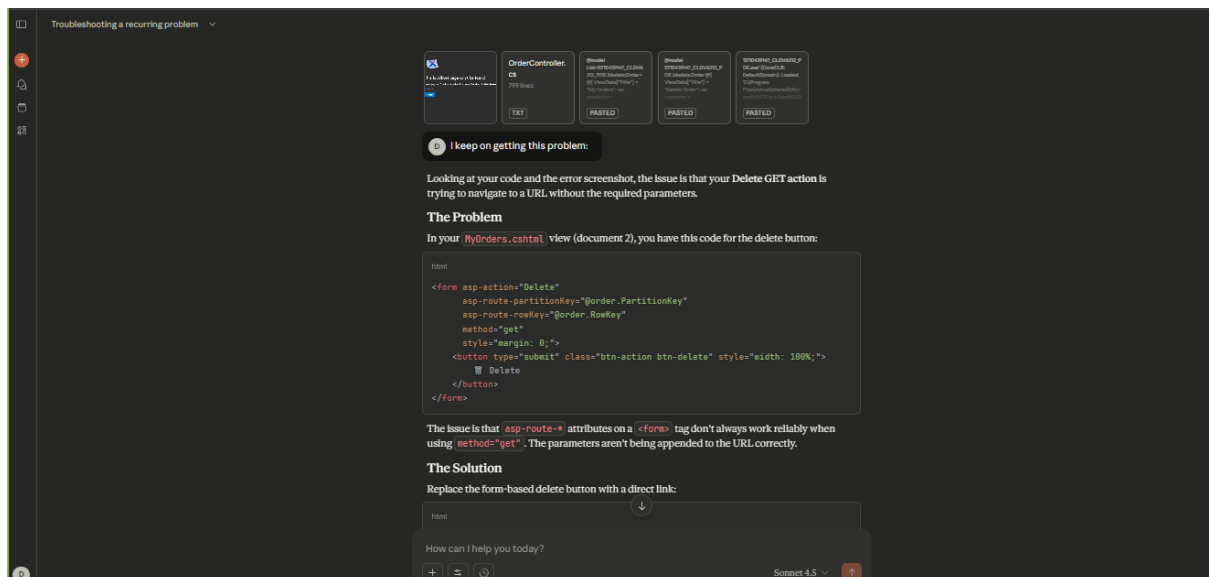
Disclosure of AI Usage in my Assessment

Coding Part

Used it to help identify why I am getting this error and identify where it is within the code.

14/11/2025

Claud.AI: <https://claude.ai/chat/5b0f17e7-5de2-41e9-83ea-2b10483de6ec>

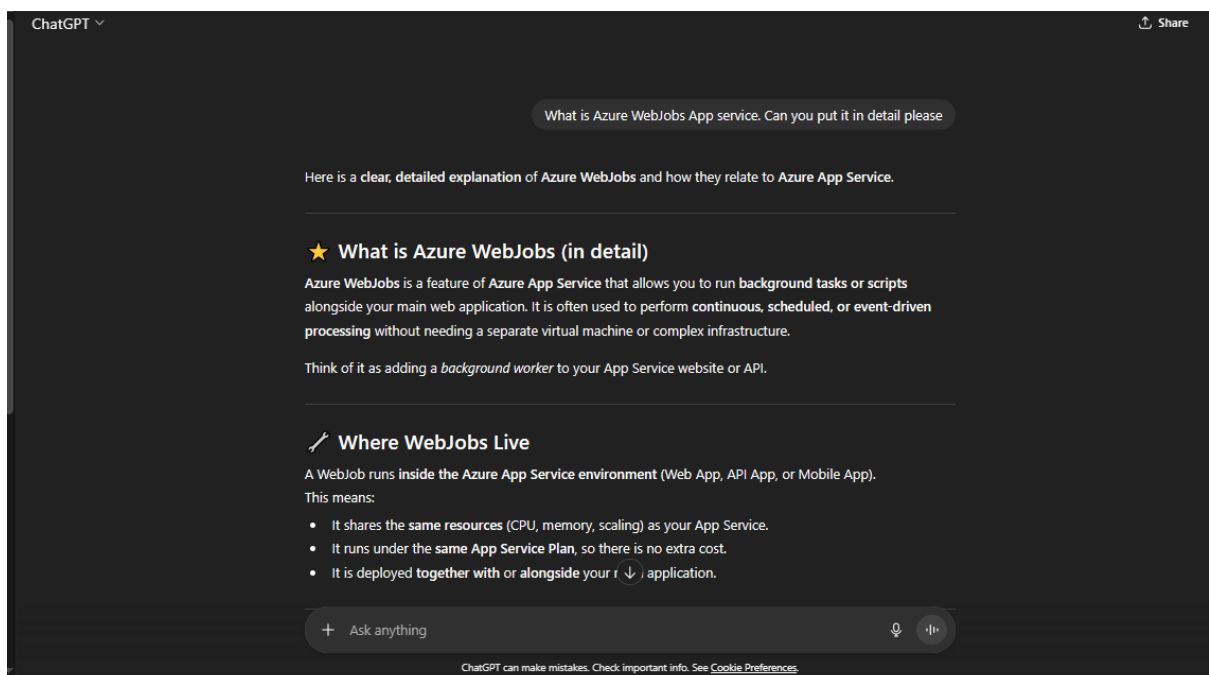


Section D

Used to help me research and see what I need to look at for Azure WebJobs App Service.

14/11/2025

ChatGPT: <https://chatgpt.com/c/6916e686-ad9c-832b-b033-6485544951b9>

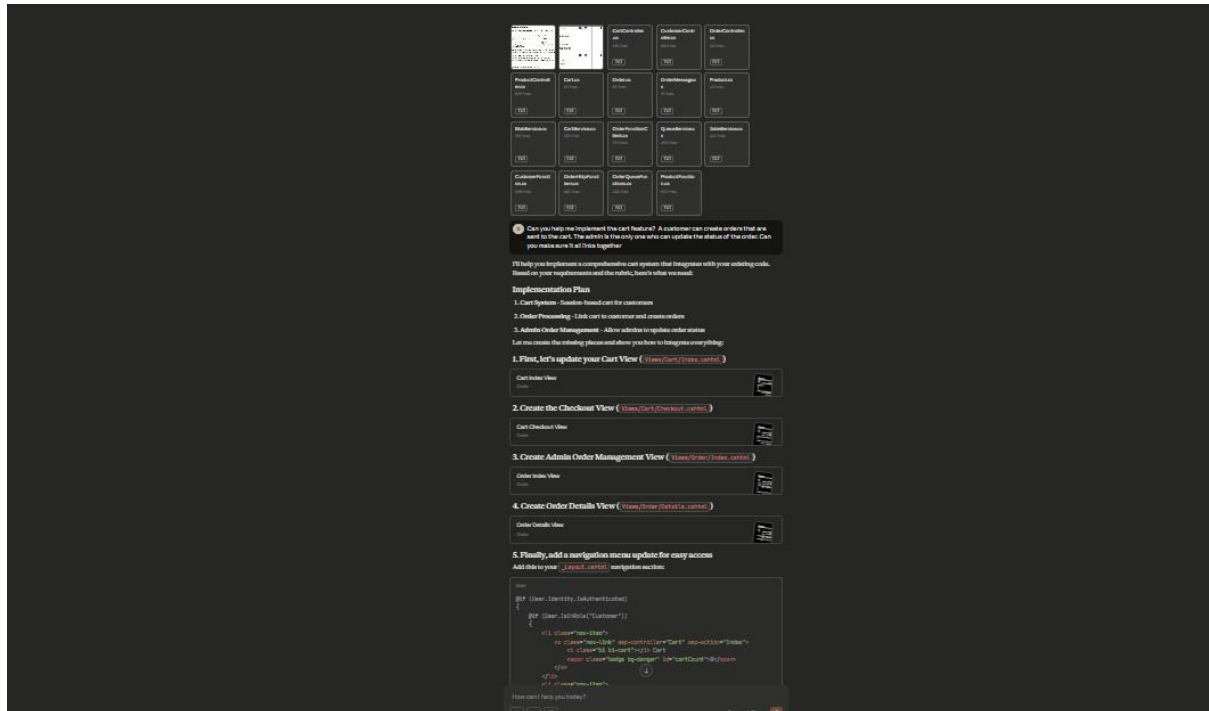


Coding part

Help me decide how I want to implement the cart feature and how I can implement it.

13/11/2025

Claud.AI: <https://claude.ai/chat/d976e455-64cc-41f4-a240-0cfdb3969ce9>



Coding part

Help me retrieve the password's Hash in order for the password to be saved correctly.

12/11/2025

ChatGPT: <https://chatgpt.com/c/6914ab8f-afa4-8332-b866-707e78f4ab88>

